

AER1217 Lab 4: Stereo Camera-based Visual Odometry

Group 2:

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1 Introduction

In this lab, we implemented a **Stereo Camera-based Visual Odometry (VO)** pipeline using images from the KITTI dataset. Our main tasks:

- **RANSAC outlier rejection** for 3D correspondences,
- **Scalar-weighted point cloud alignment** (via SVD) for each frame-to-frame pose,
- Comparison of our **VO trajectory** to the provided **ground truth**.

2 Formulations and Implementation Details

2.1 Stereo Triangulation

First, we match features among {prev-left, prev-right, cur-left, cur-right}. Each left-right pair is converted to 3D using:

$$Z = \frac{f B}{(x_L - x_R)}, \quad X = \frac{(x_L - c_u) Z}{f}, \quad Y = \frac{(y_L - c_v) Z}{f}.$$

2.2 RANSAC with 3D Alignments

We use the model:

$$\mathbf{p}_b^j = \mathbf{C}_{ba} \mathbf{p}_a^j + \mathbf{r}_b^{ab}, \tag{1}$$

where $\mathbf{p}_a^j$ is the j -th 3D point in frame a , and $\mathbf{p}_b^j$ is the corresponding point in frame b . We proceed as follows:

1. **Randomly sample** 3 correspondences $(\mathbf{p}_a^j, \mathbf{p}_b^j)$.
2. **Compute** $\mathbf{C}_{ba}$ and $\mathbf{r}_b^{ab}$ by SVD (Section 2.3).
3. **Measure residuals:** $\|\mathbf{p}_b^j - (\mathbf{C}_{ba} \mathbf{p}_a^j + \mathbf{r}_b^{ab})\|$.

4. **Inlier check:** below 0.3 m threshold (our chosen tolerance).

5. **Repeat** up to 2000 times, keep the largest inlier set.

We then *re-fit* $\mathbf{C}_{ba}$ and $\mathbf{r}_b^{ab}$ with only the inliers.

2.3 Point Cloud Alignment via SVD

After determining the inlier correspondences, we must compute a rigid-body transform

$$\mathbf{p}_b^j = \mathbf{C}_{ba} \mathbf{p}_a^j + \mathbf{r}_b^{ab}.$$

We adopt the standard method (Eqs. (2)–(6)):

$$\begin{aligned} \mathbf{p}_a &= \frac{1}{w} \sum_{j=1}^J w^j \mathbf{p}_a^j, \quad \mathbf{p}_b = \frac{1}{w} \sum_{j=1}^J w^j \mathbf{p}_b^j, \quad w = \sum_{j=1}^J w^j, \\ \mathbf{W}_{ba} &= \frac{1}{w} \sum_{j=1}^J w^j (\mathbf{p}_b^j - \mathbf{p}_b) (\mathbf{p}_a^j - \mathbf{p}_a)^T, \\ \mathbf{V} \mathbf{\Sigma} \mathbf{U}^T &= \mathbf{W}_{ba}, \quad \mathbf{C}_{ba} = \mathbf{V} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \det(\mathbf{U} \mathbf{V}^T) \end{bmatrix} \mathbf{U}^T, \\ \mathbf{r}_b^{ab} &= -\mathbf{C}_{ba}^T \mathbf{p}_b + \mathbf{p}_a. \end{aligned}$$

In our implementation:

- We set the weights $w^j = 1$ (*uniform weighting*), so $\mathbf{p}_a$ and $\mathbf{p}_b$ become simple centroids.
- We construct $\mathbf{W}_{ba}$ by summing outer products $(\mathbf{p}_b^j - \mathbf{p}_b)(\mathbf{p}_a^j - \mathbf{p}_a)^T$.
- We apply `np.linalg.svd` (or equivalent) to factor $\mathbf{W}_{ba} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$.
- To ensure a *proper* rotation (no reflection), we check $\det(\mathbf{U} \mathbf{V}^T)$. If negative, we flip the sign of the last column of $\mathbf{U}$.
- Finally, we compute $\mathbf{r}_b^{ab}$ by shifting frame a 's centroid into frame b .

3 Results vs. Ground Truth

We accumulate these frame-to-frame transforms to get a global VO trajectory. Figure 1 compares our **VO trajectory** in blue with the **ground truth** in red. Numerically, we found:

- **Mean position error:** 1.117 m
- **Final frame error:** 2.039 m

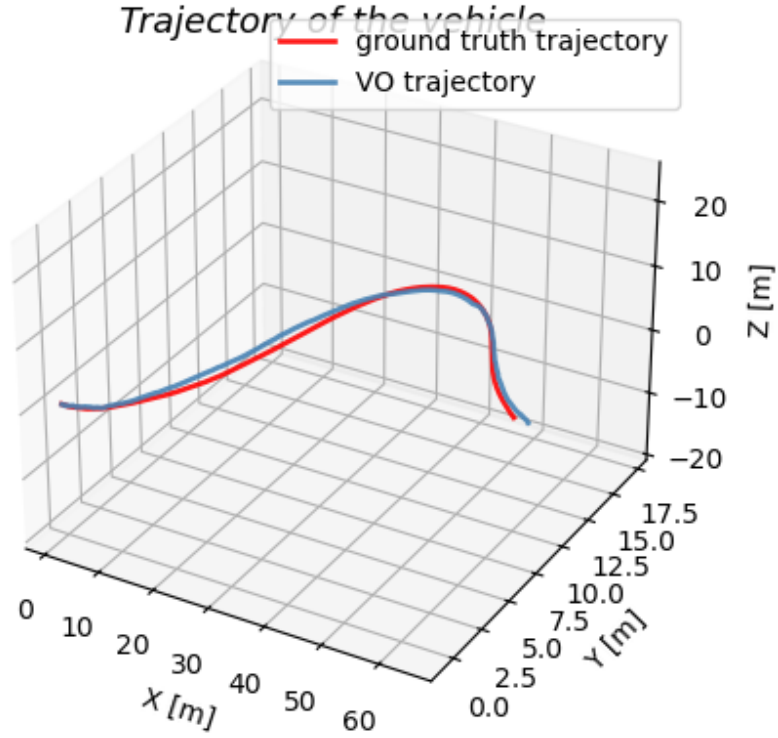


Figure 1: Ground truth (red) vs. our VO trajectory (blue).

4 Discussion

Limitations of VO:

- Drift accumulates over time, since errors in each frame's pose multiply.
- Feature-poor or dynamic scenes can reduce accurate matching.
- Large baseline rotations or occlusions can briefly break track.

Potential Improvements:

- Fuse with IMU or GPS for absolute corrections.
- Use loop closure or bundle adjustment to reduce drift.
- Tune RANSAC parameters (the 0.3 m inlier threshold or the 2000 iteration cap) for a better trade-off between speed and accuracy.

Conclusion

In summary, we designed a stereo VO pipeline that uses feature matching, outlier rejection, and an SVD-based alignment to estimate each frame-to-frame pose. Our final 3D trajectory shows close agreement with ground truth, achieving a mean position error near 1.1 m. Future directions could include IMU fusion, loop closure, or a bundle-adjustment approach to further reduce drift.